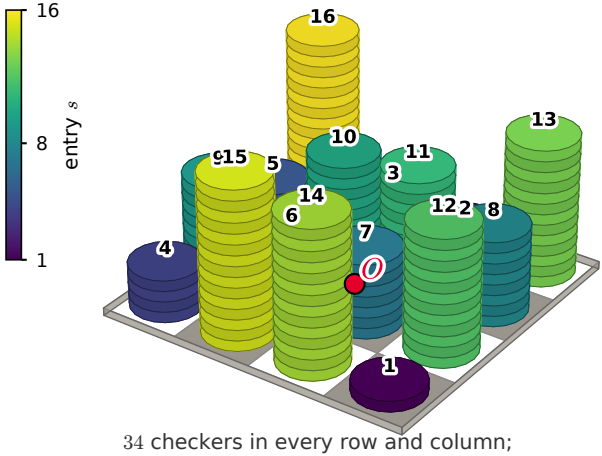
(a) checkers on a checkerboard

cell (i, j) carries a pile of s_{ij} checkers,
labelled and coloured by the entry s_{ij}

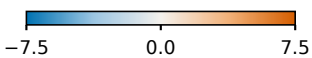


34 checkers in every row and column;
their centre of mass lies over O ($\sum s\,x = \sum s\,y = 0$)

(b) heights z_{ij}

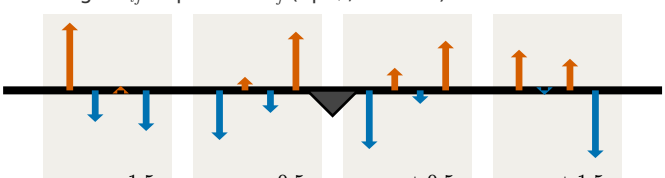
row sums ρ_i, column sums γ_j

+7.5	-5.5	-6.5	+4.5	$\rho_0 = 0$
-3.5	+1.5	+2.5	-0.5	$\rho_1 = 0$
+0.5	-2.5	-1.5	+3.5	$\rho_2 = 0$
-4.5	+6.5	+5.5	-7.5	$\rho_3 = 0$
$\gamma_0 = 0$	$\gamma_1 = 0$	$\gamma_2 = 0$	$\gamma_3 = 0$	



(c) the heights as forces on a lever

arrow = height z_{ij} at position x_j (up $+$, down $-$)



$$\sum z\,x = \sum_j x_j \gamma_j = 0$$

each column balances,
so the whole lever balances

(d) the mixed moment $\sum z\,x\,y$ is not forced: two magic squares

square S

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

$z\,x\,y$ per cell

-17	+4	-5	+10
+3	0	+1	0
0	-1	0	-3
-10	+5	-4	+17

$$\sum z\,x\,y = 0$$

square S

1	6	11	16
8	15	2	9
12	3	14	5
13	10	7	4

$z\,x\,y$ per cell

+17	+2	+2	+17
0	-2	-2	0
+3	-1	-1	+3
+10	+1	+1	+10

$$\sum z\,x\,y = +60$$